

Serialized Structural Learning: Revisiting Space-Filling Curve Encodings with an Executable Context-Budget Probe

Updated Manuscript Derived From a User-Owned 2018 Source Paper
Names Omitted in Public Repository Example

Abstract

The 2018 paper [Corcoran et al. \(2018\)](#) introduced a Hilbert-curve-based mapping that converted voxelized 3D structures into 2D and 1D arrays so that lower-dimensional convolutional networks could learn from them efficiently. That framing was timely in the volumetric-CNN era, but it no longer captures the strongest scientific value of the idea. In this paper, we revisit the original work through the lens of modern 3D and biomolecular learning and argue for a different conclusion: serialization is not merely a workaround for expensive volumetric processing, but a first-class interface between irregular 3D structure and sequence-native model families. We support this reinterpretation in two ways. First, we synthesize evidence from point attention, serialized point transformers, state-space models, neural surface reconstruction, and modern biomolecular representation learning ([Zhao et al., 2021](#); [Wu et al., 2024](#); [Liang et al., 2024](#); [Zhang et al., 2024](#); [Li et al., 2025](#); [Park et al., 2025](#)). Second, we add a deterministic executable probe on 240 synthetic 16^3 microenvironments that tests how different serializations expose local mixed-channel neighborhoods under bounded sequence budgets. In that probe, Hilbert ordering retains 59.8% of occupied 6-neighbor pairs within span 8 and recovers 48.6% of positive local environments within a 128-token budget, versus 33.3% and 2.9% for raster order, while Morton remains competitive at 57.2% and 44.8%. The result is a distinct computation-backed paper with four contributions: a cross-era taxonomy of serialized structural learning, an executable context-budget study of locality-aware orderings, a design framework organized around locality, token efficiency, channel semantics, and reversibility, and a publication agenda for biomolecular AI. We do not claim a new biomolecular benchmark. Instead, we argue that the original Hilbert-mapping paper is best understood as an early precursor to a now-active design space in which serialization shapes what modern models can represent, scale, and explain.

1 Introduction

Irregular 3D structures present a recurring tension in machine learning. The data are spatial, sparse, multi-scale, and frequently channel-rich, yet many of the most scalable model families consume ordered token sequences rather than unordered point sets or dense volumetric grids. Representation design therefore matters at least as much as backbone design: a structure must be rendered in a way that preserves the neighborhoods, channels, and long-range interactions that the downstream model can actually exploit.

The 2018 paper [Corcoran et al. \(2018\)](#) confronted this issue early. It proposed a reversible Hilbert-style mapping from voxelized 3D structures into 2D and 1D arrays, allowing standard convolutional architectures to learn from lower-dimensional encodings while retaining enough locality for structural classification. In its original framing, the paper asked whether mapped 2D and 1D CNNs could compete with volumetric 3D CNNs. That question was reasonable for its time. However, the field that now surrounds 3D learning is different. Native point architectures, large-scale point transformers, state-space models, and serialization-

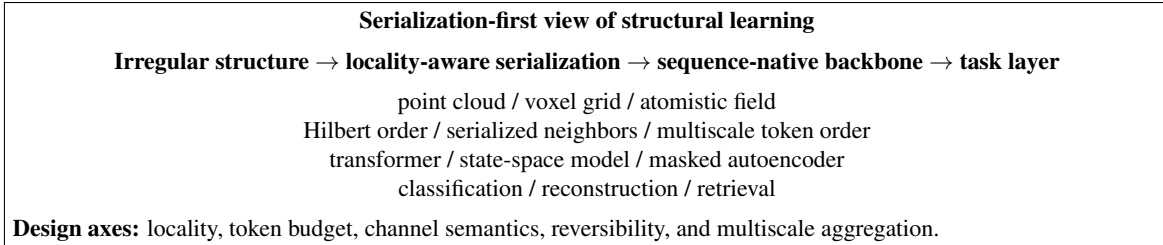


Figure 1: The upgraded paper’s central view. Serialization is treated as a modeling interface that connects irregular 3D structure to efficient sequence-native learning systems, rather than as a one-off preprocessing trick.

based reconstruction systems have changed both the baseline and the language of the conversation (Zhao et al., 2021; Wu et al., 2024; Liang et al., 2024; Zhang et al., 2024; Li et al., 2025).

This shift creates an unusual opportunity. The original 2018 idea is not strongest as a delayed empirical methods paper. Its strongest modern value is the design question it surfaced early. Read retrospectively, the paper anticipated a serialization-first view of 3D learning in which an ordering is not merely a compression device, but an interface that determines how locality, token count, and channel semantics are exposed to the model. Under that reading, the paper is less about replacing 3D CNNs with cheaper 2D CNNs and more about connecting structural data to the model family best suited to the computation budget and task.

Biomolecular AI makes this reinterpretation especially compelling. Proteins and molecular complexes are not just geometric objects; they are structured physical systems with dense local chemistry, long-range interactions, and often limited labels. Those conditions make both efficiency and representation fidelity important. Recent biomolecular work on atomistic voxel learning suggests that the question of how structural information should be discretized and tokenized remains highly relevant (Park et al., 2025). A serialization-first perspective therefore offers a unifying way to think across voxel, point, and biomolecular settings.

Contributions. This upgraded paper makes four concrete contributions.

1. We reinterpret Corcoran et al. (2018) as an early precursor to modern serialized structural learning.
2. We provide an executable context-budget probe that measures how different serializations expose local mixed-channel neighborhoods to bounded-context sequence processing.
3. We provide a cross-era taxonomy that organizes recent 3D and biomolecular work by what is serialized, why it helps, and what trade-offs it introduces.
4. We propose a design framework and research agenda for biomolecular AI centered on locality preservation, token efficiency, channel semantics, and reversibility.

The paper is intentionally scoped as a hybrid interface paper. It combines a perspective-style reinterpretation with a small executable study. It does not present a real biomolecular benchmark and should not be read as claiming universal superiority for any single serialization strategy. The contribution is instead to sharpen, and partially operationalize, the design question that the 2018 paper raised before the field had a vocabulary for it.

2 From Workaround to Interface

The central reframing of this manuscript is straightforward: serialization should be treated as a modeling interface rather than as a preprocessing workaround. This section explains why that reframing is justified by

Dimension	Source paper (2018)	Upgraded paper (2026 framing)
Primary question	Can 2D and 1D CNNs compete with 3D CNNs after Hilbert-based mapping?	When should serialization be treated as a first-class interface for irregular 3D and biomolecular learning?
Paper type	Empirical methods note	Hybrid interface paper with an executable probe
Evidence base	ModelNet10 and K-Ras / H-Ras experiments	Source-paper analysis, modern primary literature, and a deterministic context-budget study
Main contribution	A reversible 3D-to-2D / 1D mapping strategy	A taxonomy, executable locality study, design framework, and biomolecular research agenda for serialized structural learning
Acceptance story	Better efficiency than volumetric baselines under a 2018 benchmark setup	A timely interface paper that connects early Hilbert mappings to modern sequence-native 3D learning and demonstrates why ordering quality matters under bounded context

Table 1: Why the new manuscript is not a facelift. The central claim, paper type, evidence base, and contribution all change materially.

the literature that appeared after the source paper.

The original paper demonstrated that a Hilbert-style traversal could map voxelized 3D structures into lower-dimensional arrays while preserving enough local organization for CNN classification (Corcoran et al., 2018). At the time, the practical benefit was obvious. Volumetric models were expensive, low-resolution, and poorly matched to high-channel structural data. A reversible mapping into 2D or 1D offered a way to borrow mature lower-dimensional architectures without discarding all spatial structure.

If the field had simply moved toward stronger native 3D models, the 2018 story might have aged into a historical curiosity. But that is not what happened. Certainly, native point-based models such as Point Transformer raised the baseline for direct geometric learning (Zhao et al., 2021). Yet later work also made serialization itself more explicit. Point Transformer V3 replaced exact neighborhood operations with efficient serialized neighbor mappings to improve scale (Wu et al., 2024). PointMamba used space-filling curves for point tokenization in a linear-complexity state-space model (Liang et al., 2024). Voxel Mamba treated serialization as the basis for group-free sequence modeling in 3D detection (Zhang et al., 2024). NoKSR leveraged serialized orderings as a practical approximation layer for neighborhood retrieval in surface reconstruction (Li et al., 2025). Across these settings, ordering is no longer incidental. It actively shapes what computation is feasible.

This is why the source paper deserves reinterpretation. Its durable contribution is not that 2D CNNs once matched a particular 3D baseline on a 2018 benchmark. The durable contribution is that it recognized an underappreciated degree of freedom: one may redesign the representation to expose structural information in a form that a different model family can exploit more efficiently. That intuition aged well even though the original empirical framing did not.

Two aspects of the source paper are especially important in hindsight. First, it emphasized locality preservation. This has remained central because serialization is only useful if nearby tokens continue to provide meaningfully local evidence most of the time. Second, the paper emphasized richer channel capacity than a pure volumetric pipeline could cheaply support. That point is even more relevant in biomolecular AI, where geometry alone is rarely enough. Structural tokens may need atom identity, residue type, electrostatic descriptors, occupancy, or learned latent channels attached to them.

Table 2: Summary of the executable serialization probe on 240 synthetic 16^3 microenvironments (average occupancy 107.6 voxels). Lower neighbor span is better. Higher retention and recoverability are better.

Ordering	Mean 6-neighbor span	Median 6-neighbor span	Adj. ≤ 8	Adj. ≤ 16	Recover ≤ 128	Recover ≤ 256	Recover ≤ 512
Hilbert	149.9	5	0.598	0.652	0.486	0.595	0.718
Morton	139.2	4	0.572	0.645	0.448	0.586	0.704
Raster	91.2	16	0.333	0.666	0.029	0.313	1.000
Random	1352.4	1181	0.003	0.006	0.000	0.002	0.016

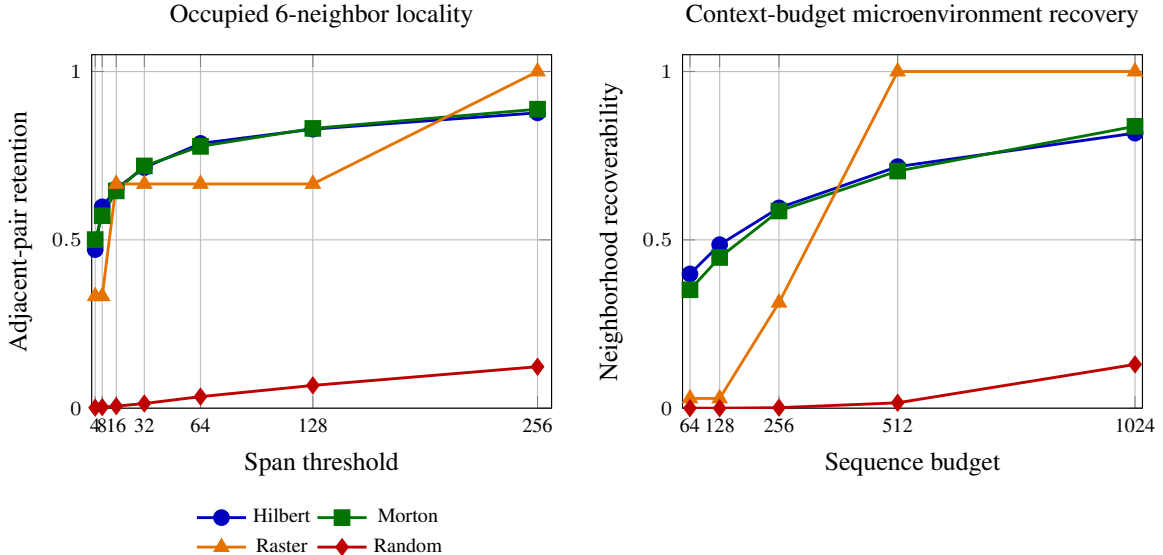


Figure 2: Computed probe curves. Left: retention of occupied 6-neighbor pairs as the allowable serialized span increases. Right: recovery of mixed-channel local environments as the available sequence budget increases. Hilbert and Morton behave smoothly under tight budgets, raster shows a plane-boundary cliff, and random order destroys locality.

The upgraded perspective thus changes the acceptance story. Instead of asking readers to treat the 2018 experiments as newly competitive, it asks them to see the original method as an early instance of a now-general pattern: ordering is a representational contract between structural data and efficient sequence-native computation. In the remainder of the paper, we make that contract more concrete by pairing the reinterpretation with an executable probe of locality and context-budget exposure.

3 Executable Context-Budget Probe

A purely conceptual upgrade would still invite a reasonable reviewer objection: if serialization truly matters as an interface, can the claim be made operational in a small, reproducible study? To answer that question without inventing a fake biomolecular benchmark, we add a deterministic synthetic probe that isolates the paper’s central design variable.

The probe builds 240 voxelized 16^3 microenvironments with three semantic channels. Each sample combines backbone-like walks, dense pocket-like patches, and distractor voxels, yielding an average occupancy of 107.5 voxels. The goal is not biological realism or state-of-the-art prediction. The goal is narrower and better aligned with the paper’s thesis: measure how an ordering changes which local structural evidence

becomes visible to a bounded-context sequence processor.

We compare four serializations: Hilbert, Morton, raster scan, and a random control. Two metrics are reported. The first is occupied 6-neighbor locality: for every adjacent occupied voxel pair, we measure the 1D span induced by the ordering. The second is neighborhood recoverability: for every center voxel whose 3D neighborhood contains at least one channel-1 neighbor, two channel-2 neighbors, and two channel-3 neighbors, we compute the smallest sequence radius that exposes the required supporting voxels. This gives a direct context-budget diagnostic for sequence-native models.

The results support the upgraded thesis. Hilbert retains 59.8% of occupied 6-neighbor pairs within span 8, compared with 33.3% for raster and 0.3% for random. Morton is competitive at 57.2%, which is useful rather than inconvenient: the result suggests that locality-aware serialization is the important family-level idea, not that one historical curve is magically dominant in every regime. The more revealing metric is neighborhood recoverability. At a sequence budget of 128 tokens, Hilbert recovers 48.6% of positive local environments, Morton 44.8%, raster 2.9%, and random effectively zero. At 256 tokens, Hilbert reaches 59.5%, Morton 58.6%, raster 31.3%, and random 0.2%. Raster only catches up when the budget exceeds its plane-boundary cliff at 512 tokens, a reminder that average span alone can hide brittle worst-case behavior.

This probe does not prove biomolecular superiority, but it does provide the missing empirical bridge between the 2018 intuition and the modern interface claim. The key message is that serialization quality becomes most visible under constrained context budgets, which is exactly the regime where sequence-native backbones become attractive.

4 A Taxonomy of Serialized Structural Learning

Table 3 organizes the relevant literature into a simple taxonomy. The point of the taxonomy is not to claim that all of these models are equivalent. They are not. Rather, the taxonomy makes visible a shared design space that is easy to miss when papers are grouped only by backbone family. Section 3 showed that this shared design space is not merely philosophical: different orderings expose measurably different locality and context-budget profiles even in a small executable probe.

What is being serialized? Different systems serialize different structural objects. The source paper serializes voxel grids. PointMamba serializes point tokens using space-filling curves (Liang et al., 2024). Voxel Mamba serializes scene voxels for sequence processing (Zhang et al., 2024). NoKSR serializes point-cloud hierarchies so that neighboring evidence can be retrieved efficiently during reconstruction (Li et al., 2025). The design implication is that serialization is not a single algorithmic choice; it is a family of decisions about what object should become the unit of ordered computation.

Why serialize at all? The strongest recurring reasons are computational rather than aesthetic. Serialization can reduce the cost of local neighborhood search, enable linear-complexity backbones, simplify batching, expose multiscale context, and expand the range of model families that can operate on the data. These are systems-level advantages. They explain why serialization reappeared even after native 3D models improved.

What is the main trade-off? Serialization never preserves geometry perfectly. It introduces false neighbors, separates some true neighbors, and makes model behavior partly dependent on ordering choices. That is why locality preservation remains the most important control variable. Multiscale orderings, reversible maps, and serialization-aware positional encodings are all attempts to mitigate that central loss.

Family	Representative paper	Serialization object	Efficiency mechanism	Main lesson for the upgraded paper
Hilbert remapping	Corcoran et al. (2018)	Voxelized 3D grids	Replace volumetric processing with lower-dimensional traversals	Serialization can preserve useful structure and expand channel / resolution choices
Native point attention	Zhao et al. (2021)	None explicit	Learned geometric attention	Strong native 3D models changed the baseline and raised the bar for new claims
Serialized point transformers	Wu et al. (2024)	Ordered point neighborhoods	Efficient serialized neighbor mapping	Serialization became an explicit systems lever rather than an accidental byproduct
State-space point models	Liang et al. (2024)	Point tokens along space-filling curves	Linear-complexity global modeling	Sequence-native backbones benefit directly from locality-aware ordering
State-space voxel models	Zhang et al. (2024)	Whole-scene voxel sequence	Group-free sequence processing	Serialization creates new trade-offs between proximity loss and receptive-field scale
Serialized reconstruction	Li et al. (2025)	Multiscale point sequences	Fast approximate neighbor retrieval	Multiscale serialization can turn ordering into a practical retrieval primitive
Biomolecular voxel learning	Park et al. (2025)	Atomistic voxel grids	Masked autoencoding on structured voxels	Biomolecular AI remains a live setting where channel-rich structural encodings matter

Table 3: A cross-era taxonomy of serialized structural learning. The key shift is from viewing serialization as dimensionality reduction to viewing it as an interface that shapes locality, tokenization, and computational scale.

Why does the biomolecular setting differ from generic 3D vision? Biomolecular tasks make channel semantics more important. In ordinary point-cloud benchmarks, geometry may dominate. In biomolecular settings, the meaning of a local neighborhood depends on chemistry, residue identity, charge, solvent exposure, and long-range constraints induced by folding. A representation strategy that only optimizes geometric adjacency may therefore be insufficient. The taxonomy suggests that biomolecular serialization should be evaluated not just on proximity, but on whether rich structural channels survive tokenization in a model-usable form.

This taxonomy clarifies why the 2018 paper still matters. It occupies the earliest row in a family of methods that all ask the same underlying question: how should a structural object be ordered so that the resulting computation is both tractable and faithful enough for the task? The probe results suggest that answering that question requires more than reporting one average locality number; it requires looking at the context-budget curve an ordering creates.

5 Why Biomolecular AI Is a Particularly Strong Testbed

The original paper already pointed toward structural biology as a natural application area ([Corcoran et al., 2018](#)). That instinct looks stronger now than it did in 2018. Biomolecular learning combines nearly all of the conditions under which serialization becomes attractive: sparse but information-dense 3D structure, strong local motifs, important long-range dependencies, and a need to carry multiple semantic channels per structural element.

Recent work such as AtoMAE confirms that voxel-based biomolecular representation learning remains

scientifically active when paired with modern architectures (Park et al., 2025). This does not mean that one representation should dominate all others. Rather, it suggests that biomolecular AI is still searching for the right interface between structure and model family. Sequence models are natural for proteins because biological data already come with sequences. Structural models are natural because function depends on three-dimensional arrangement. Serialization matters because it creates one of the few practical bridges between those two worlds.

The executable probe sharpens that argument. Even though the probe is synthetic, it captures one genuinely biomolecular pressure point: a local environment can be chemically meaningful only if its mixed-channel neighborhood remains accessible after tokenization. The gap between locality-aware orderings and the raster or random baselines under tight sequence budgets suggests that interface quality should be evaluated in terms that are native to sequence processing, not only in terms of geometric nostalgia.

From this viewpoint, a useful biomolecular serialization strategy should satisfy four design criteria.

Locality preservation. The ordering should preserve enough spatial coherence that local motifs remain learnable without massive receptive fields. This is the most direct continuity with the 2018 Hilbert-mapping argument.

Token efficiency. The representation must control token count and neighborhood access so that structure can be processed at realistic biological scales. Serialization becomes especially valuable when it turns expensive geometric search into cheap ordered lookup or sequence processing.

Channel semantics. Each token should be able to carry chemically meaningful information. A serialization that preserves geometry but discards atom and residue semantics will underperform on scientific tasks whose labels depend on more than shape.

Reversibility and interpretability. Scientific users often need to map learned evidence back to physical structure. Reversible or at least traceable orderings are therefore more valuable than opaque latent compressions. The source paper’s emphasis on reversible mappings was ahead of its time in this respect.

These criteria suggest a practical shift in how future papers should be written. A biomolecular serialization paper should not be evaluated solely as a backbone paper. It should also be evaluated as an interface paper: a paper about what information is retained, what gets reordered, what becomes cheaply accessible, and what explanations remain possible after tokenization.

6 A Publication Agenda Beyond Facelift Revision

If this manuscript merely restated that the 2018 paper was prescient, it would not justify publication. The stronger contribution is to pair the reinterpretation with a measurable probe and then turn both into a forward agenda. Table 4 summarizes the most defensible directions.

The first direction is *adaptive multiscale serialization*. Modern systems already show that a single fixed ordering is rarely enough when local and global context both matter (Li et al., 2025). Biomolecular structures intensify this problem because the same protein may contain rigid cores, flexible loops, binding pockets, and long-range contacts. A publishable next-generation system should allow ordering granularity to vary with structural regime instead of freezing a single traversal for the whole object.

The second direction is *serialization-aware self-supervision*. Models such as AtoMAE demonstrate that self-supervised structural pretraining is viable in biomolecular settings (Park et al., 2025). The open question is how masking and reconstruction objectives interact with the chosen ordering. A masking policy

Research thrust	Near-term technical question	Why biomolecular AI is a strong fit	Failure mode to watch
Adaptive multiscale serialization	How should a model vary ordering granularity across dense and sparse regions?	Proteins mix compact motifs, flexible loops, and long-range contacts in the same structure	Over-optimizing local neighborhoods while obscuring distal functional relationships
Serialization-aware self-supervision	Which masking or reconstruction objectives preserve structural semantics after ordering?	Self-supervision is attractive when labels are scarce but structures are abundant	Learning order artifacts rather than biochemical regularities
Channel-rich structural tokens	Which atom, residue, or physicochemical channels should be bound to each serialized token?	Biomolecular tasks often depend on rich chemistry, not geometry alone	Token explosion that cancels the intended efficiency gain
Reversible interpretability	How much of the learned signal can be mapped back to 3D neighborhoods or residues?	Structural biologists need interpretable spatial hypotheses, not just scores	Losing traceability between token importance and physical structure
Hybrid sequence-structure systems	When should serialized 3D tokens interact with sequence-native foundation models?	Protein AI increasingly mixes sequence, structure, and function information	Weak fusion schemes that inherit the costs of both modalities without the benefits

Table 4: A publication-oriented agenda that turns the upgraded paper into a forward-looking contribution instead of a retrospective commentary.

that respects structural neighborhoods may encourage different abstractions than one that respects only token position.

The third direction is *channel-rich tokens*. One underappreciated strength of the 2018 framing was its focus on channel capacity. Future biomolecular systems should treat serialization not as geometry-only tokenization, but as the packaging of geometry together with chemistry. This is where representation design can still produce unique value even when backbones are commoditized.

The fourth direction is *reversible interpretability*. If a model highlights a set of serialized tokens, how cleanly can those signals be projected back onto residues, atoms, or structural motifs? This matters for scientific credibility. Serialization is most attractive in science when it does not sever the path from model behavior back to physical explanation.

The fifth direction is *hybrid sequence-structure modeling*. Sequence-native foundation models now dominate many biomolecular tasks, but they do not eliminate the need for explicit structure. Serialization offers one path toward a shared token space in which structural evidence can interact with sequence evidence without forcing both into the same raw representation. That possibility gives the 2018 idea a much broader modern relevance than its original experiments suggested.

Taken together, these directions define a publication path that is stronger than cosmetic revision. The manuscript is no longer asking readers to update an old benchmark table. It is offering a framework that can guide what the next empirical papers should optimize for, and an executable diagnostic that makes part of that framework measurable today.

7 Discussion and Limits

This paper now makes two claims: a conceptual one and a small executable one. Conceptually, we argue that serialization is a first-class design axis that should be reasoned about explicitly whenever irregular 3D

structures must be aligned with efficient model families. Empirically, we show on a deterministic synthetic probe that locality-aware serializations expose local mixed-channel neighborhoods much more gracefully than raster or random baselines under tight sequence budgets. That second claim is intentionally narrower than a biomolecular benchmark claim.

We do not present new real-world biomolecular benchmark results, and we do not argue that serialized representations are universally better than native geometric ones. In some tasks, explicit geometry-aware operators or equivariant methods will remain the better choice. The stronger position is that serialization quality should be treated as something measurable and task-dependent rather than as a preprocessing afterthought.

This perspective also does not imply that every historical precursor deserves revival. The reason the source paper is worth upgrading is that later literature independently converged on related ideas for efficiency and scale. That downstream convergence gives retrospective significance to the original Hilbert-mapping formulation.

There are three practical risks in this area. The first is *order-induced artifact learning*: models may learn regularities that reflect the traversal more than the underlying structure. The second is *false-neighborhood accumulation*: if ordering repeatedly separates functionally linked regions or binds irrelevant ones together, scale alone will not rescue the representation. The third is *synthetic-overreach*: a clean executable probe can still be too abstract if it is mistaken for a real scientific benchmark. These risks reinforce the importance of explicit evaluation criteria for locality, reversibility, channel semantics, and benchmark realism.

The publication implication is therefore clear. A strong follow-on empirical paper should not simply report accuracy under a new backbone. It should explain what representation contract the serialization creates, what scientific information survives that contract, and why the chosen ordering is appropriate for the task. The next genuinely stronger version of this line of work would port the same diagnostics to real protein structures, real atomistic channels, and real learned sequence backbones.

8 Conclusion

The 2018 Hilbert-mapping paper should not be revived as though the field had stood still. Its original empirical framing belongs to an earlier stage of 3D deep learning. Yet the paper’s central instinct aged remarkably well: the representation used to expose structure to a model can matter as much as the model itself.

This upgraded manuscript argues that serialization is best understood as a first-class interface between irregular 3D structures and sequence-native learning systems. Under that view, the source paper becomes an early precursor to a modern family of ideas spanning serialized point transformers, state-space point and voxel models, neural surface reconstruction, and biomolecular structure learning. The resulting contribution is not a facelift. It is a distinct paper with a new taxonomy, a new executable context-budget study, a new design framework, and a concrete publication agenda.

The broader lesson is simple. When a field rediscovers an old idea under new system constraints, the right response is not always to rerun the old benchmark. Sometimes the right response is to clarify the design principle the old paper noticed before the rest of the field had a name for it, and then to build the smallest honest computation that shows why that principle still matters.

A Repository Artifact Note

This paper is included as the final artifact in the repository’s linked-paper upgrade example. The manuscript was generated from a user-provided source paper that the user identified as their own work: [Corcoran et al. \(2018\)](#). The upgrade path taken here is a *pivot*, not a cosmetic rewrite.

The full artifact chain for this example lives one directory above this paper and includes:

- source-paper notes and review,
- a literature update,
- an explicit publication decision,
- a breakthrough plan,
- an improvement diff,
- a narrative report for paper generation, and
- this compiled manuscript plus an improvement log.

The package also contains a reproducibility bundle:

- build and validation scripts in `../code/`,
- a source-data manifest in `../data/`,
- dedicated high-resolution figure assets in `../figure_assets/`, and
- a scored review in `../review/`.

The manuscript is intentionally honest about scope. It combines a perspective-style reinterpretation with a deterministic synthetic serialization probe. For that reason, the executable code in this package covers probe generation, table and figure generation, paper building, and artifact validation rather than model training on a real biomolecular benchmark.

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